INFORMATION TECHNOLOGY

CONTINGENCY PLAN

Emergency Preparedness and Preventive Measures for Information Technology Disasters

UNION PETROCHEMICAL PUBLIC COMPANY LIMITED

(INFORMATION TECHNOLOGY CONTINGENCY PLAN)

Information is an important asset of the Company and must be safeguarded to ensure security and effective use in business operations. Emergency incidents arising from internal or external factors may damage databases, information systems and equipment.

Accordingly, the IT Department has prepared this Information Technology Contingency Plan as a framework for maintaining, protecting and resolving issues affecting the Company's information technology, covering:

1) Analysis and assessment of disaster severity
2) Preventive measures and initial preparedness
3) Organizational structure and assignment of responsibilities during emergencies
4) Disaster prevention and response measures
5) Disaster recovery and restoration to normal operations
6) Monitoring and reporting

Emergency Response Plan Management and Risk Management

1. Analysis and Assessment of Disaster Severity

1.1 Disaster Analysis

Disasters that may cause damage to information technology are classified into two groups: External Disasters

1) Natural disasters and civil disturbances affecting the building housing servers and main processing equipment, such as floods, fires, earthquakes, humidity and temperature-related incidents.
2) Theft of server equipment used for data storage and collection.
3) Failure of server communication systems connected to the internet.
4) Electrical system failure, power outages or unstable power supply.
5) External intrusion or attacks intended to access or control IT systems, including damage to or destruction of data.
6) Computer viruses.

Internal Disasters

1) Damage to server systems or primary databases, or destruction of data.
2) Computer viruses originating from internal users.
3) Employees' insufficient knowledge of computer hardware or software, causing IT systems to malfunction, become unusable or stop operating.

1.2 Situation Assessment and Severity Classification

Following disaster analysis, the severity of each event is assessed and classified to prepare for responses to security incidents. IT personnel shall maintain a Security Log Management System to record and analyze events and compile information for subsequent reporting.

Emergency Situations	Severity Level			Total Score	Priority Rank
	Impact on Operations	Impact on Mission	Impact on Mission		
Fire	5	5	5	15	1
System Breach	5	3	5	13	2
Power Outage	5	1	5	11	3
Flooding	4	2	4	10	4
Earthquake	4	1	5	10	5
Protests	2	3	4	9	6
Civil Unrest	2	2	4	8	7
Political Instability	2	2	4	8	8
Storm	2	1	5	8	9
Epidemic	1	1	5	7	10
Drought	1	1	4	6	11

2. Preventive Measures and Initial Preparedness

2.1 Procedures

The IT Department shall prepare procedures for handling abnormal events. When an emergency occurs, the procedures appropriate to the circumstances shall be selected to collect incident information, identify the source of intrusion or terminate the incident accurately and promptly. Backup equipment shall be available for restoring critical systems when problems arise.

2.2 Communication

A contact list of relevant external agencies shall be maintained for security-related communication in emergencies, including the Provincial Electricity Authority, police stations and fire services. Coordination with the building security center shall include building plans and internal access routes for the fire services (see the attached contact list referenced in the original document).

2.3 Preparation of Essential Equipment

As the central unit responsible for computer services and support, the IT Department shall prepare the following equipment and tools for computer failures and IT-related disasters:

1) Installation media for operating systems, network operating systems and critical applications.
2) Backup tapes for data and critical systems.
3) Antivirus/spyware software installation media.
4) Device driver installation media.
5) Emergency backup power systems.
6) Other relevant spare computer equipment.

2.4 Data Backup

To prevent data loss, destruction, unauthorized alteration or damage caused by computer viruses or intruders, backups shall enable affected data to be restored to normal operation. The Company has established a Backup and Information Technology Continuity Plan Policy within the Veeam Backup system (spelled “Veem Backup” in the source).

2.5 Computer Virus

Prevention Antivirus software shall be installed on servers and client computers connected to the network. Users must exercise care when operating computer systems, particularly when connecting to the internet, to preventunauthorized access or system damage. Under the Virus and Malicious Software Protection Policy:

1) A Fortigate firewall is installed and utilities are kept updated to defend against virus attacks.
2) Virus protection is installed on servers.
3) All client computers use licensed operating systems with Windows Defender antivirus protection.

2.6 Prevention and Resolution of Electrical Power Problems

1) Install uninterruptible power supplies (UPS) to prevent damage from voltage drops and surges; replace degraded batteries to maintain power storage capability.
2) UPS backup power for servers provides approximately 15–30 minutes of operation.
3) Keep the UPS switched on while computers are in use and perform maintenance.
4) Ensure backup power equipment remains ready for use.
5) In a power outage, users shall immediately save work in progress and shut down computers and other equipment.

2.7 Prevention of Computer

Intrusions and Threats The following measures are intended to strengthen the security of information systems and computer networks:

1) Restrict access to the network control room to authorized personnel. Where access is necessary, responsible IT personnel shall supervise entry and exit. CCTV is installed to deter theft.
2) Install a firewall to prevent unauthorized access from the internet.
3) Keep the firewall enabled at all times to protect information systems and computer networks.
4) Use the PRTG network monitoring system to continuously monitor incidents and produce written reports.

5) Install a proxy server to improve internet service performance and filter website data; configure it securely for the network and information systems.
6) Network administrators shall monitor company internet traffic for unusual volumes or unusually frequent system access, investigate causes and implement preventive measures.
7) Compliance with the Computer-Related Crime Act B.E. 2550 (2007) helps strengthen protection against computer intrusion and threats.

3. Organizational Structure and Responsibilities During Emergencies

The IT Department has established a response team with clearly assigned duties and responsibilities as follows:

3.1 Policy Level

Responsible for setting policy, providing recommendations and advice, and monitoring, supervising, controlling and reviewing operational personnel:

1) Company Board of Directors
2) Managing Director (MD)
3) IT Division/Department Manager

3.2 Operational Level:

1) Mr. Thanakorn Wongsachai — Tel. 098-5953991
2) DITC Co., Ltd. — Tel. 02-5550999

4. Disaster Prevention and Response Measures

The Company has established the following procedures for personnel to prevent and address disasters affecting its information systems.

4.1 Client Computers

1) If a computer cannot operate or access information systems normally, employees shall report the incident to the network administrator through the LINE group. IT Support will provide remote assistance within the service period specified in the Service Level Agreement (SLA).
2) If a malfunction is caused by a computer virus, the responsible person shall immediately disconnect the affected computer from the network to prevent propagation. If there is a risk to other units at the site, first disconnect the LAN cable from the relevant floor network connection point.
3) Relevant IT personnel shall investigate and perform initial troubleshooting. If the issue cannot be resolved, they shall report it to their supervisor for further action.

4.2 Servers

1) Disconnect network connections promptly and shut down network devices and servers according to service priority.
2) In the event of a power outage or voltage drop, shut down servers and network equipment according to service priority.
3) Disconnect the electrical supply. In case of fire, use fire-extinguishing agents to control the fire and inspect the incident; if conditions are unsafe, promptly move equipment to a safe location.
4) In case of fire, promptly use a fire-extinguishing agent to control the flames.
5) Move equipment to a safe location as quickly as possible.
6) Promptly coordinate assistance from responsible external agencies or computer maintenance consultants.
7) If hardware fails, promptly obtain replacement equipment or ask the responsible service company to replace it as soon as possible.

5. Disaster Recovery Plan

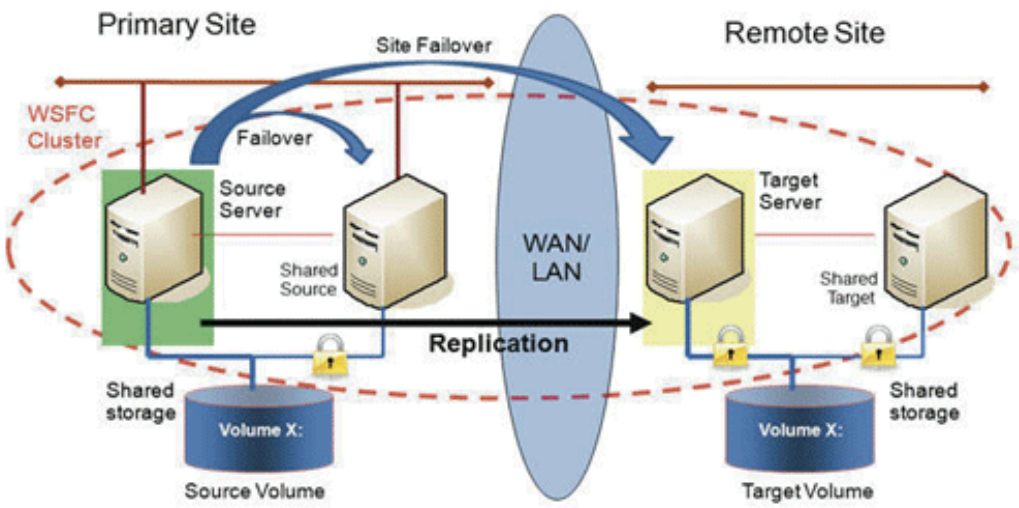
Servers and network distribution equipment should normally be available to support client computers 24 hours a day. If service becomes unavailable, recovery must be performed as quickly as possible, to the extent feasible. This plan describes how computer operations and data are restored after system damage or interruption:

1) Procure new replacement equipment.
2) Replace damaged components.
3) Complete repair and maintenance of damaged equipment within two days (48 hours).
4) Borrow computer equipment temporarily from other departments.
5) Restore backed-up data from backup tapes, CD-ROMs or hard disks through the recovery team, completing recovery as soon as possible and within 48 hours.
6) Check the operating system, database, data integrity and other related systems.

Disasters include not only hardware-related incidents such as fire, flooding, earthquakes and sabotage, but also system hacking and computer viruses affecting IT operations. The Company therefore plans to maintain backup data resources at an alternate site to ensure continuity of IT services.

Implementation Plan

1) Assess backup system requirements.
2) Identify an appropriate alternate site.
3) Assess risks and establish risk management measures.
4) Prioritize organizational impacts.
5) Prepare the recovery plan.
6) Plan the response team and sequence of activities following system damage.
7) Train personnel on their responsibilities and provide technical training.
8) Test the plan.
9) Improve the system recovery plan.



Benefits of the Disaster Recovery Plan

1) Raise awareness of security and potential impacts.
2) Reduce the risk of delays during incident resolution.
3) Provide assurance of the reliability of backup systems.
4) Establish standards for crisis response plans and testing.
5) Reduce decision-making time during crises.
6) Reduce potential legal liabilities.
7) Reduce unnecessary stress in the working environment.

6. Monitoring and Reporting

Responsible personnel shall report operational or inspection results to their supervisor, namely the IT Division/Department Manager. The manager shall report these results to the Managing Director and Board of Directors on a monthly basis. Problems and corrective actions shall also be reported immediately once action can be taken in all cases described above. The reports shall be used to improve and develop the IT security plan so that it remains effective and can be implemented promptly in the event of future disasters.

Signature: 

(Mr. Perapol Suwannapasri)

Managing Director